Benefits management system - Integrative software engineering, semester B, 2024

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Project summary document- 17\7\24

1. Introduction:

The City Library System is designed to streamline operations and enhance user experience at the city library. The system includes three main applications: the Patron Application for library users, the Librarian Dashboard for library staff, and the Room Reservation System for booking study rooms. The vision is to create a comprehensive, user-friendly platform that integrates library management, book borrowing, and room reservations into a seamless digital experience.

1.1 Purpose of System

The primary business purpose of the City Library System is to improve the efficiency of library operations and enhance user satisfaction. This system aims to:

- Facilitate easy access to library resources for patrons.
- To Enable efficient management of library inventory and tracking the status of books and manage users' information
- To provide a convenient way to reserve and manage room bookings.

1.2 Scope of System

In Scope:

- Development of the Patron Application for book searching and borrowing
- Creation of the Librarian Dashboard for managing books and patrons
- Implementation of the Room Reservation System for booking and managing room reservations.
- Integration with the library’s database to ensure real-time updates and data consistency.

Beyond Scope:

- Integration with external libraries or inter-library loan systems: The system will support direct integration with other libraries' databases and allow inter-library loans.
- Advanced analytics and reporting tools beyond basic usage statistics: The system will include sophisticated data analysis tools to provide deeper insights into user behavior and library performance.
- Mobile application development for platforms other than the primary web-based system: The project will include the development of separate mobile applications for Android and iOS.
- Maintenance and support for outdated browser versions or operating systems: The system will support older versions of browsers and operating systems to ensure compatibility.
- Personalized recommendations: The system will provide personalized book recommendations based on user behavior and preferences.
- Automated reminders: The system will send automated reminders for book due dates and room reservation times.
- Multi-language support: The system will be developed to support multiple languages to cater to a diverse user base.

2. Actors and Goals

Actors:

Patron (Primary)

Description: Library users who search for, borrow, as well as reserve rooms.

Objectives:

- Search for and locate books.
- Borrow books.
- Reserve study and meeting rooms.

Librarian (Primary)

Description: Library staff responsible for managing the library's book inventory, patron records.

Objectives:

- Add, update, and remove books from the library inventory, tracking books
- Manage patron records, including registration and updates.

Administrator (Support)

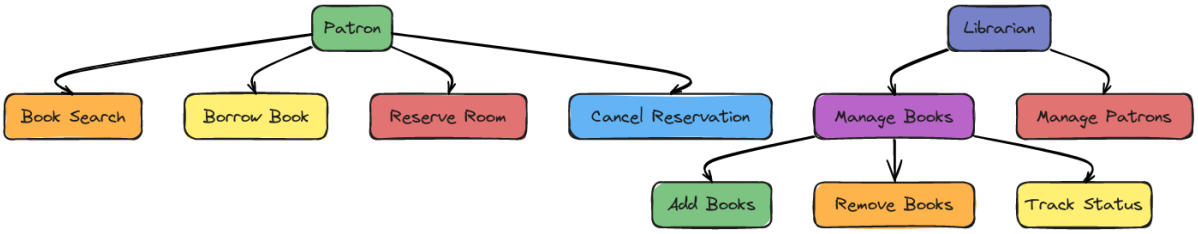
Description: System administrators who oversee the overall operation of the library system and handle system configurations and maintenance.

Goals:

- Ensure system reliability and performance.
- Manage user roles and permissions.
- Oversee database backups and security measures.

USE CASES

Use Case Diagram:



Use Case Details :

Patron Application

Use Case: Book Search

Use Case: Display Book Search Option

Name: Display Book Search Option
Goals: Allow patrons to select the book search option.
Participating actors: Patron

Basic Workflow:

1. The patron navigates to the library homepage.
2. The patron sees the "Search For Book" option.

Alternate Workflow:

- None

Use Case: Enter Book Search Details

Name: Enter Book Search Details
Goals: Allow patrons to enter book title, author, or genre in search fields.
Participating actors: Patron

Basic Workflow:

1. The patron selects the "Search For Book" option.
2. The patron enters the book title, author, or genre in the search fields.

Alternate Workflow:

- None

Use Case: Perform Book Search

Name: Perform Book Search
Goals: Allow patrons to search for books in the database.
Participating actors: Patron

Basic Workflow:

1. The patron clicks the "Search" button.
2. The system searches the entire database for matching books.

Alternate Workflow:

- **Alternate Flow 1**: If no matching results are found, the system displays a message indicating that no results were found.
    - **Point of Divergence**: After step 2 in the Basic Workflow.

Use Case: Display Search Results

Name: Display Search Results
Goals: Display the search results to the patron.
Participating actors: Patron

Basic Workflow:

1. The system displays the search results to the patron.

Alternate Workflow:

- None

Use Case: Borrow Book

Use Case: Display Borrow Book Option

Name: Display Borrow Book Option
Goals: Allow patrons to select the borrow book option.
Participating actors: Patron

Basic Workflow:

1. The patron navigates to the library homepage.
2. The patron sees the "Borrow Book" option.

Alternate Workflow:

- None

Use Case: Enter Book Title to Borrow

Name: Enter Book Title to Borrow
Goals: Allow patrons to enter the title of the book they want to borrow.
Participating actors: Patron

Basic Workflow:

1. The patron selects the "Borrow Book" option.
2. The patron enters the title of the book they want to borrow.

Alternate Workflow:

- None

Use Case: Check Book Availability

Name: Check Book Availability
Goals: Check if the book is available for borrowing in the database.
Participating actors: Patron

Basic Workflow:

1. The patron clicks the "Search" button.
2. The system checks the availability of the book in the library database.

Alternate Workflow:

- **Alternate Flow 1**: If the book is not available for borrowing, the system does not display the book in the search results.
    - **Point of Divergence**: After step 2 in the Basic Workflow.

Use Case: Confirm Book Borrowing

Name: Confirm Book Borrowing
Goals: Confirm the borrowing of the book and update its status.
Participating actors: Patron

Basic Workflow:

1. If the book is available for borrowing, the system reserves the book for the patron and updates the book's status to "borrowed."
2. The system confirms the borrowing to the patron.

Alternate Workflow:

- None

Librarian Dashboard

Use Case: Book Management

Name: Manage Books
Goals: Allow librarians to manage the library's book inventory.
Participating actors: Librarian

Sub-Use Case: Add Books

- **Name**: Add Book
- **Goals**: Allow librarians to add new books to the library database.
- **Participating actors**: Librarian
- **Basic Workflow**:
    - The librarian selects "Manage the Books" option.
    - The librarian selects "Add Books to the Library."
    - The librarian enters the book details (title, author, genre, etc.).
    - The librarian clicks the "Add Book" button.
    - The system adds the book to the library database and confirms the addition.
- **Alternate Workflow**:
    - If there is an error in adding the book, the system displays an appropriate error message.

Sub-Use Case: Remove Books

- **Name**: Remove Book
- **Goals**: Allow librarians to remove books from the library database.
- **Participating actors**: Librarian
- **Basic Workflow**:
    - The librarian selects "Manage the Books" option.
    - The librarian selects "Remove Books from the Library."
    - The librarian searches for the book to be removed.
    - The librarian selects the book and clicks the "Remove Book" button.
    - The system removes the book from the library database and confirms the removal.
- **Alternate Workflow**:
    - If there is an error in removing the book, the system displays an appropriate error message.

Sub-Use Case: Track Status of Books

- **Name**: Track Book Status
- **Goals**: Allow librarians to view the current status of books in the library.
- **Participating actors**: Librarian
- **Basic Workflow**:
    - The librarian selects "Manage the Books" option.
    - The librarian selects "Track Status of Books."
    - The librarian views the current status (available, borrowed, reserved) of books in the library.
- **Alternate Workflow**:
    - If there is an error in tracking the status of books, the system displays an appropriate error message.

Use Case: Patron Management

Name: Manage Patrons
Goals: Allow librarians to manage patron information.
Participating actors: Librarian
Basic Workflow:

1. The librarian selects "Manage the Patrons" option.
2. The librarian can view and update patron information.
3. The system updates the library database with the changes.

- **Alternate Workflow**:
    - If there is an error in updating patron information, the system displays an appropriate error message.

Room reservation

Use Case: Reserve Room

Name: Display Available Rooms
Goals: Show patrons a list of available rooms for reservation.
Participating actors: Patron

Basic Workflow:

1. The patron navigates to the room reservation page.
2. The system displays a list of available rooms.

Alternate Workflow:

- None

Use Case: Select Room to Reserve Name: Select Room to Reserve
Goals: Allow patrons to select a room from the list of available rooms.
Participating actors: Patron

Basic Workflow:

1. The patron selects a room from the list of available rooms.

Alternate Workflow:

- None

Use Case: Confirm Room Reservation Name: Confirm Room Reservation
Goals: Confirm the reservation of a selected room and update its status.
Participating actors: Patron

Basic Workflow:

1. The patron clicks the "Reserve" button.
2. The system confirms the reservation and updates the room's status to "reserved."

Alternate Workflow:

- **Alternate Flow 1**: If the room is already reserved, the system displays a message indicating that the room has already been reserved.
    - **Point of Divergence**: After step 1 in the Basic Workflow.

Use Case: Cancel Reservation

Name: Display Existing Reservations
Goals: Show patrons a list of their existing room reservations.
Participating actors: Patron

Basic Workflow:

1. The patron navigates to the list of their reservations.
2. The system displays the list of reservations.

Alternate Workflow:

- None

Use Case: Select Reservation to Cancel Name: Select Reservation to Cancel
Goals: Allow patrons to select a reservation they want to cancel.
Participating actors: Patron

Basic Workflow:

1. The patron selects the reservation they want to cancel from the list of reservations.

Alternate Workflow:

- None

Use Case: Confirm Reservation Cancellation Name: Confirm Reservation Cancellation
Goals: Confirm the cancellation of a selected reservation and update the room's status.
Participating actors: Patron

Basic Workflow:

1. The patron clicks the "Cancel" button.
2. The system confirms the cancellation and updates the room's status to "available."

Alternate Workflow:

- None

4. Non Functional Requirements

Requirement Number	Requirement Type	Requirement Description
NF-001	U - Usability	The system shall have a user-friendly interface that is easy to navigate for all user roles.
NF-002	S - Supportability	The system shall support integration with third-party tools and APIs.
NF-003	P - Performance	The system shall handle up to 1000 concurrent users without significant performance degradation.
NF-004	R - Reliability	The system shall have an uptime of 99.9% to ensure availability and reliability for all users.

List of Commands for Each Mini-App-appendix

List of Actions for Each Mini-App

Patron Application Actions

1. **Register**: User registers a new account.
2. **Login**: User logs into the system.
3. **Search Books**: User searches for books in the library's catalog.
4. **Borrow Book**: User borrows a book from the library.

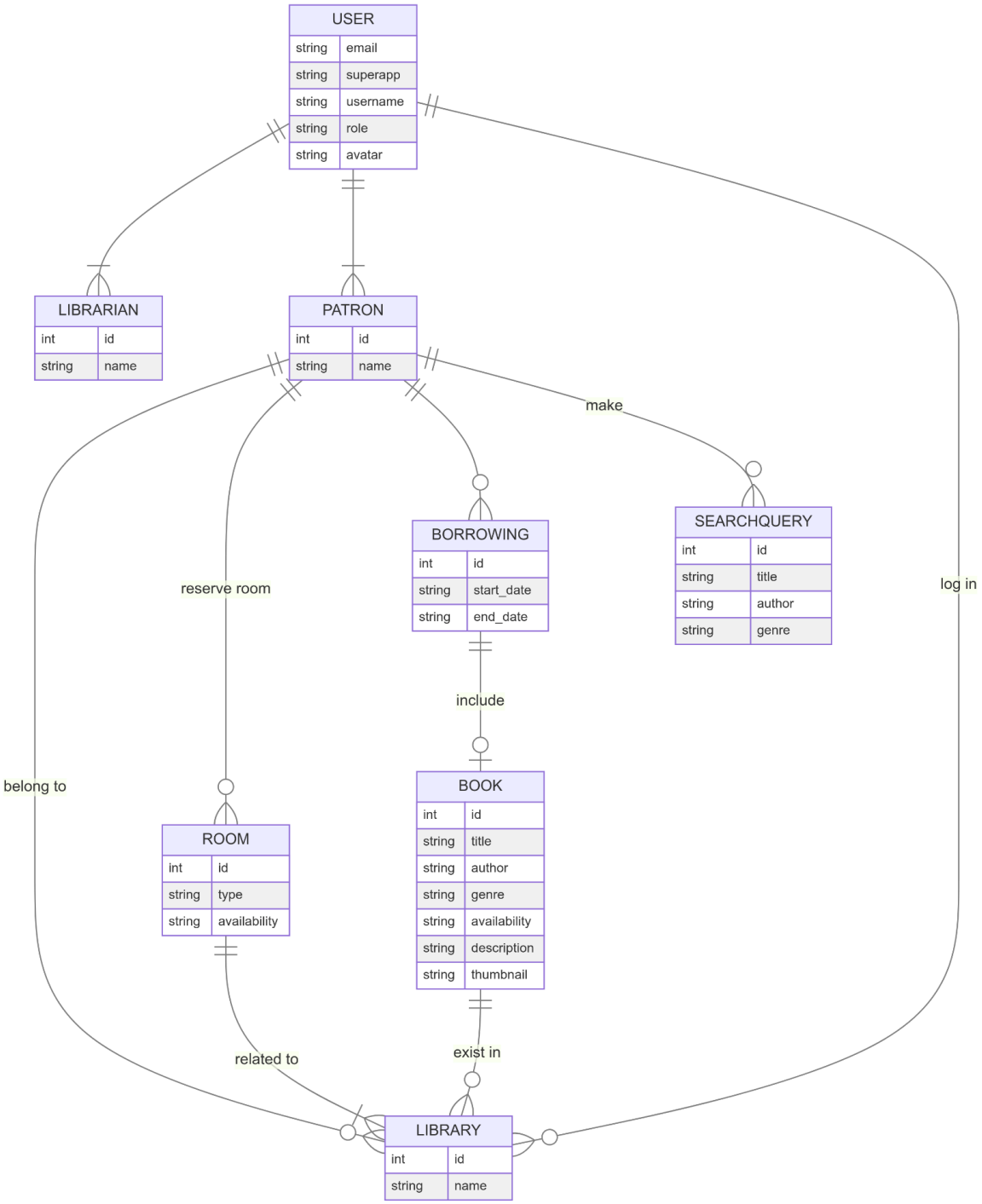
Room Reservation Actions

1. **Reserve Room**: User reserves a room in the library.
2. **View Reservations**: Users view their room reservations.
3. **Cancel Reservation**: User cancels a room reservation.

Librarian Dashboard Actions

1. **Add Book**: Librarian adds a new book to the system.
2. **Delete Book**: Librarian deletes a book from the system.
3. **Manage User Accounts**: Librarian manages user accounts.
4. **Track Book Status**: Librarian views the current status (available, borrowed, reserved) of books in the library.

UML CLASS


USER
string email
string superapp
string username
string role
string avatar
LIBRARIAN
int id
string name
PATRON
int id
string name
make
BORROWING
int id
string start_date
string end_date
SEARCHQUERY
int id
string title
string author
string genre
log in
reserve room
include
belong to
ROOM
int id
string type
string availability
BOOK
int id
string title
string author
string genre
string availability
string description
string thumbnail
related to
exist in
LIBRARY
int id
string name

technologies-appendix

Spring Framework:

- An open-source framework for building Java applications. It provides a comprehensive infrastructure for dependency injection, aspect-oriented programming, transaction management, and support for MVC. We used Spring Boot to develop and deploy our RESTful services.

Swagger:

- A tool for documenting and interacting with RESTful APIs. It provides an interactive interface for developers and users to explore and test API endpoints. We used Swagger for automatic documentation of our API interfaces.

Docker:

- A platform for running applications in containers. It allows for building, shipping, and running applications in isolated environments. We used Docker to create containers for our applications, facilitating easy and quick deployment across different environments.

H2 Database:

- A lightweight, in-memory database written in Java. It is primarily used for development and testing due to its speed and ease of use. We used H2 for our development and testing environments.

PostgreSQL:

- An advanced, open-source relational database management system (RDBMS). It offers advanced features such as transaction management, complex queries, and advanced indexing. We used PostgreSQL as the main database for our project.

React:

- A JavaScript library for building dynamic and responsive user interfaces. It allows for the creation of reusable and independent components. We used React to develop the user interfaces for our applications.

HTML, CSS, JavaScript:

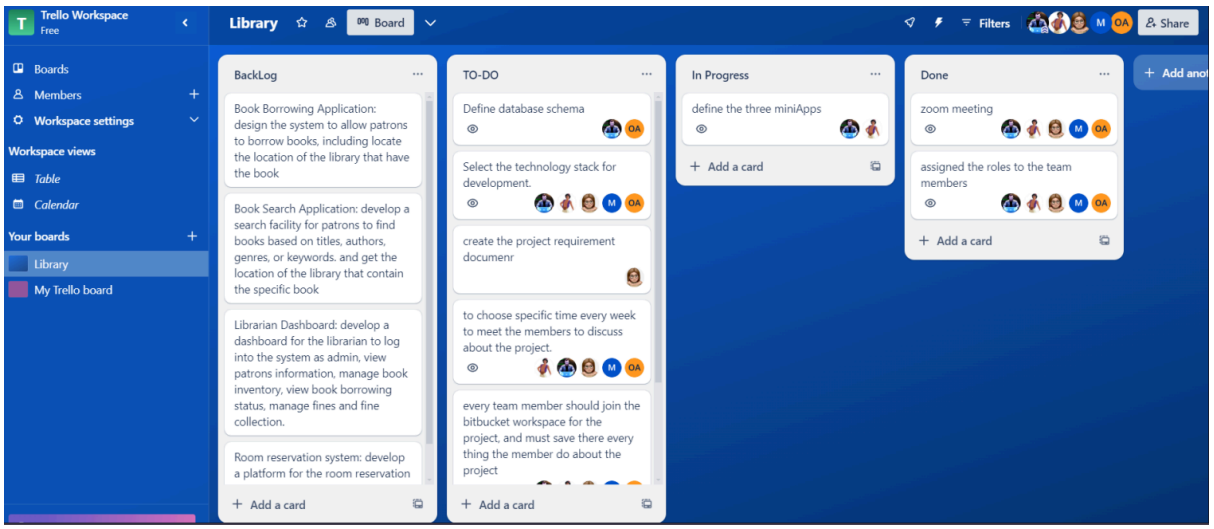
- **HTML**: The standard markup language for creating web pages.

- **CSS**: A style sheet language used for describing the presentation of a document written in HTML.
- **JavaScript**: A programming language used to create dynamic and interactive effects within web browsers.

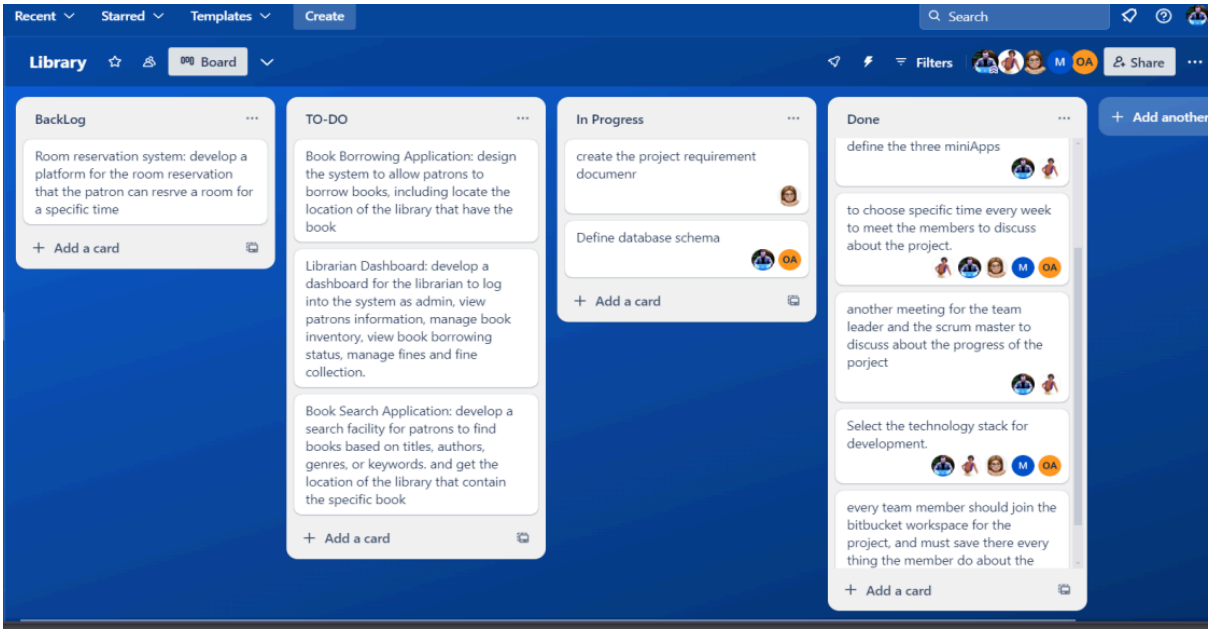
We used these technologies to build and style web pages and to develop dynamic .functionality in our user interfaces

Kanban Board

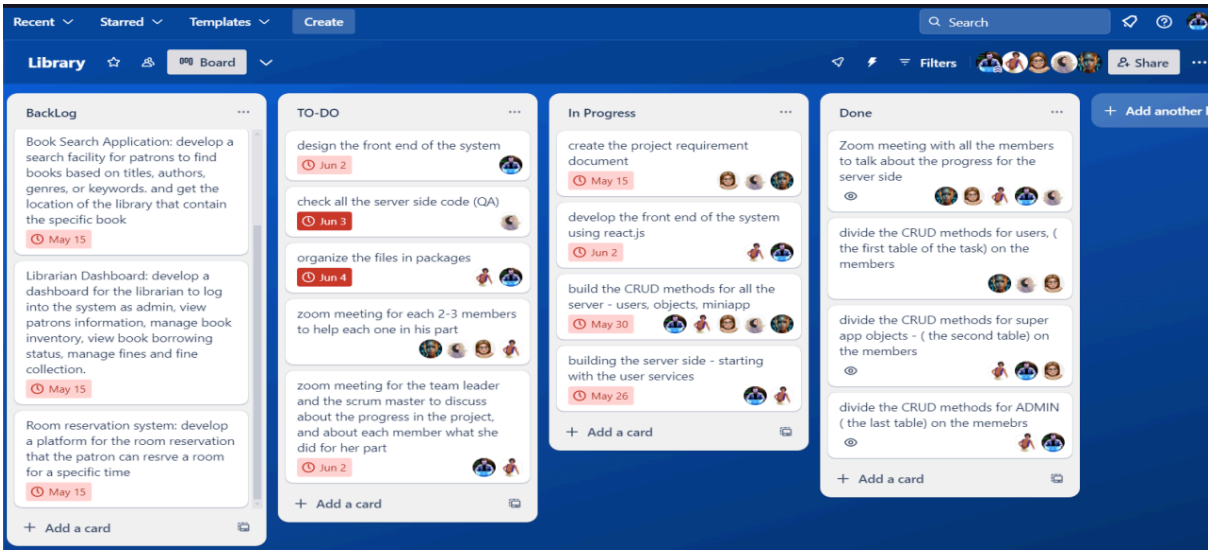
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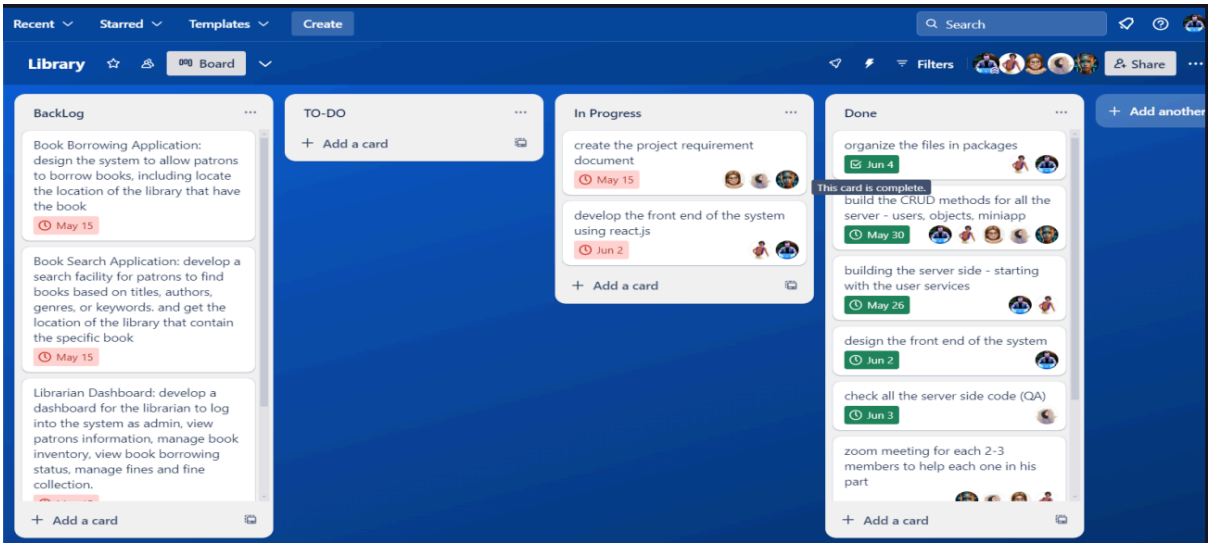
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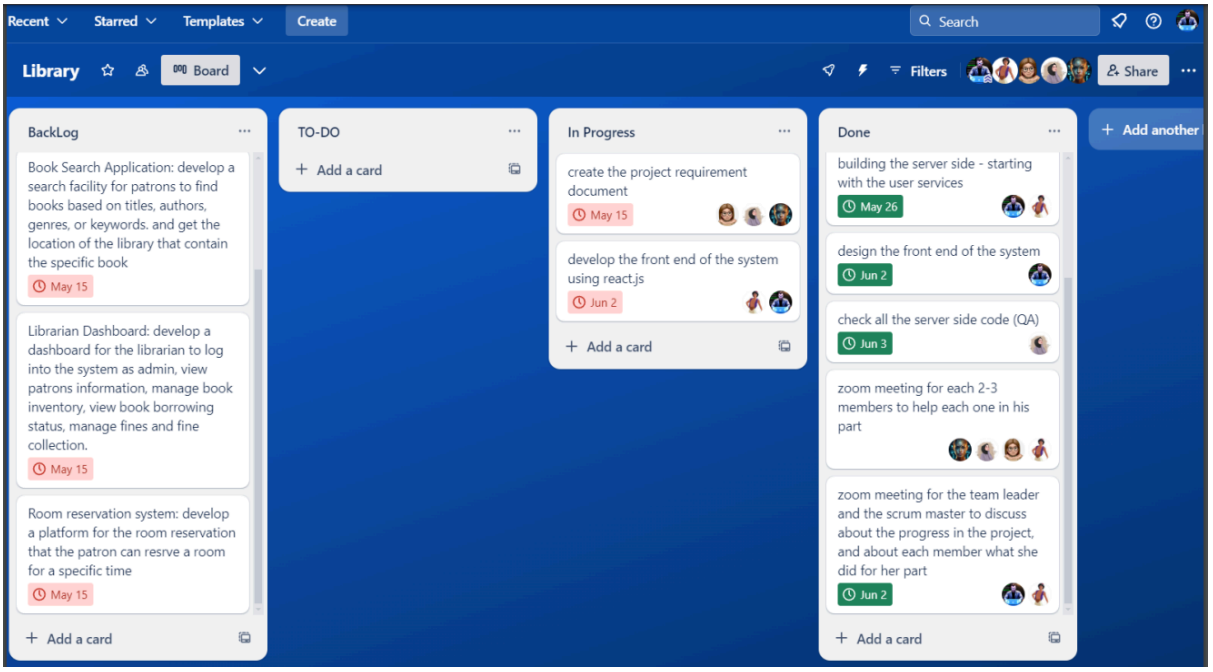


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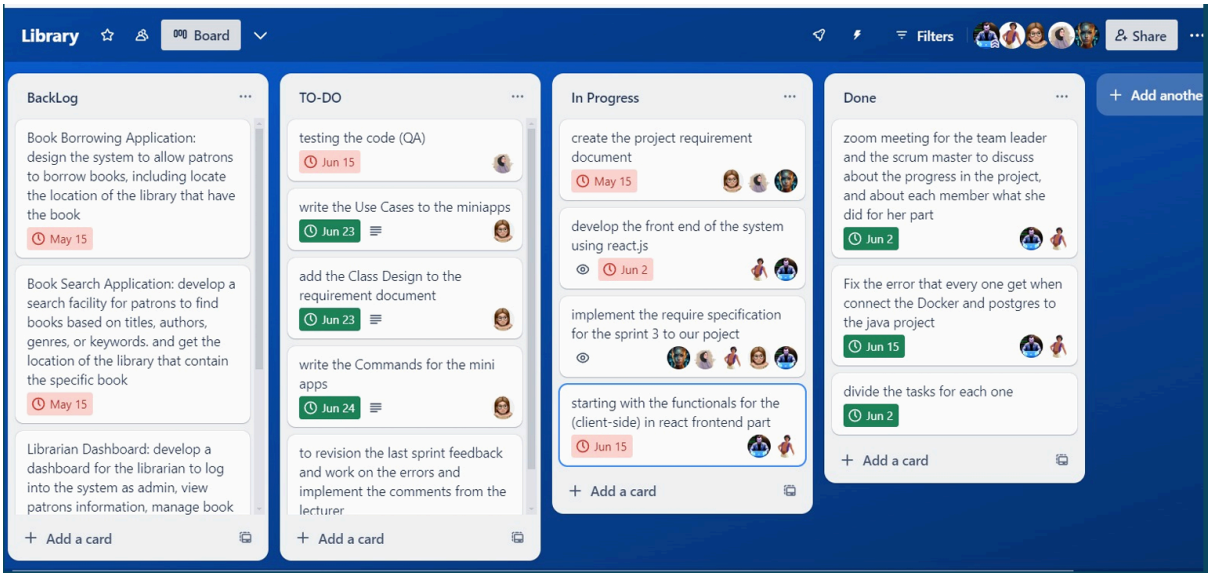


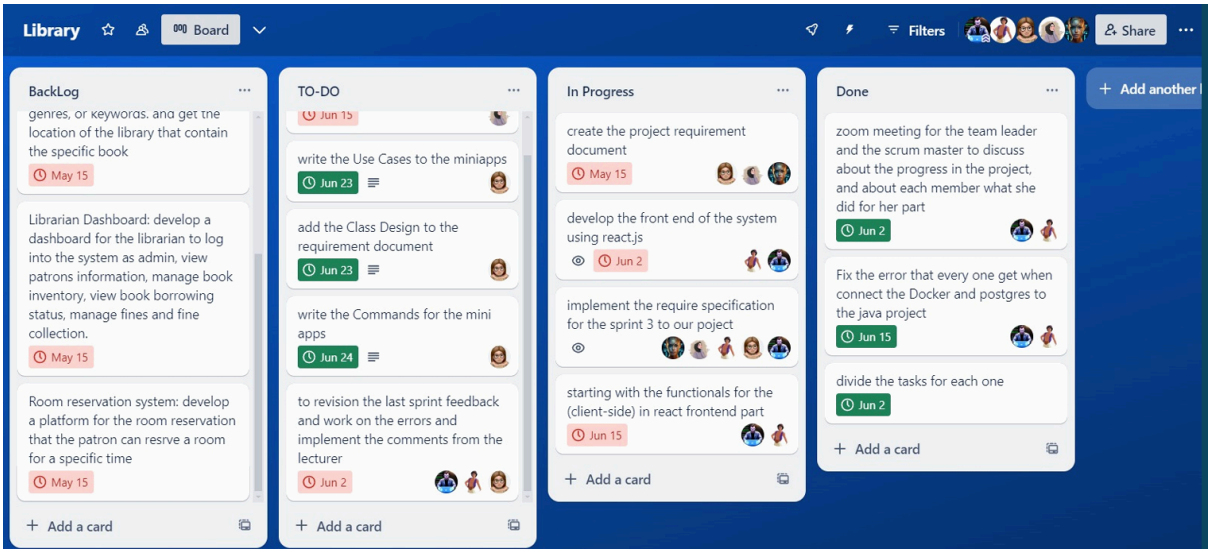
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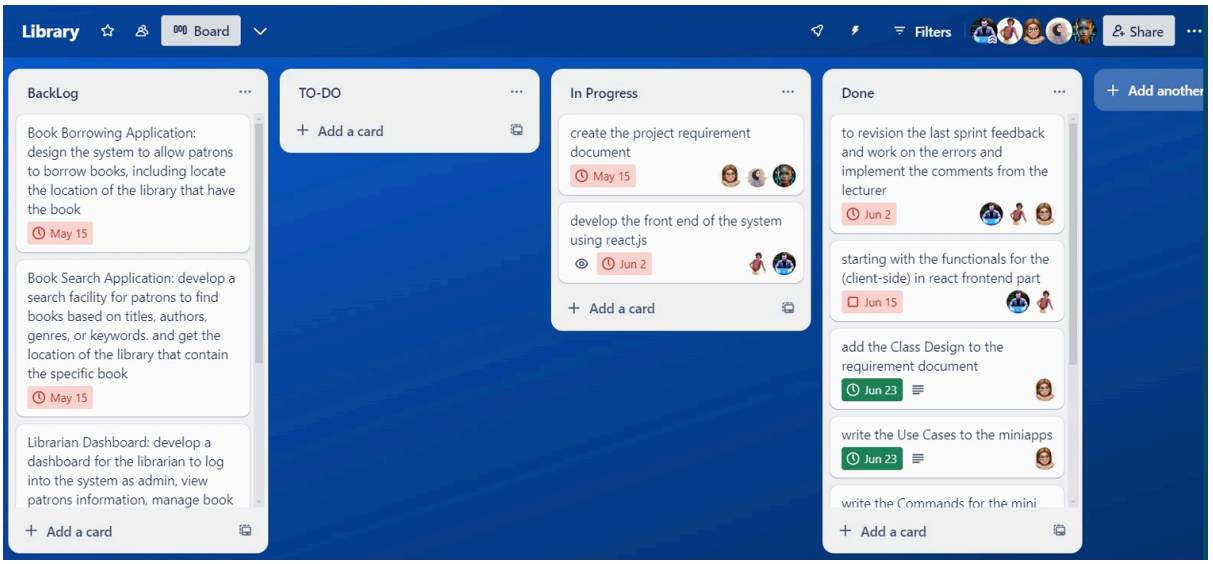


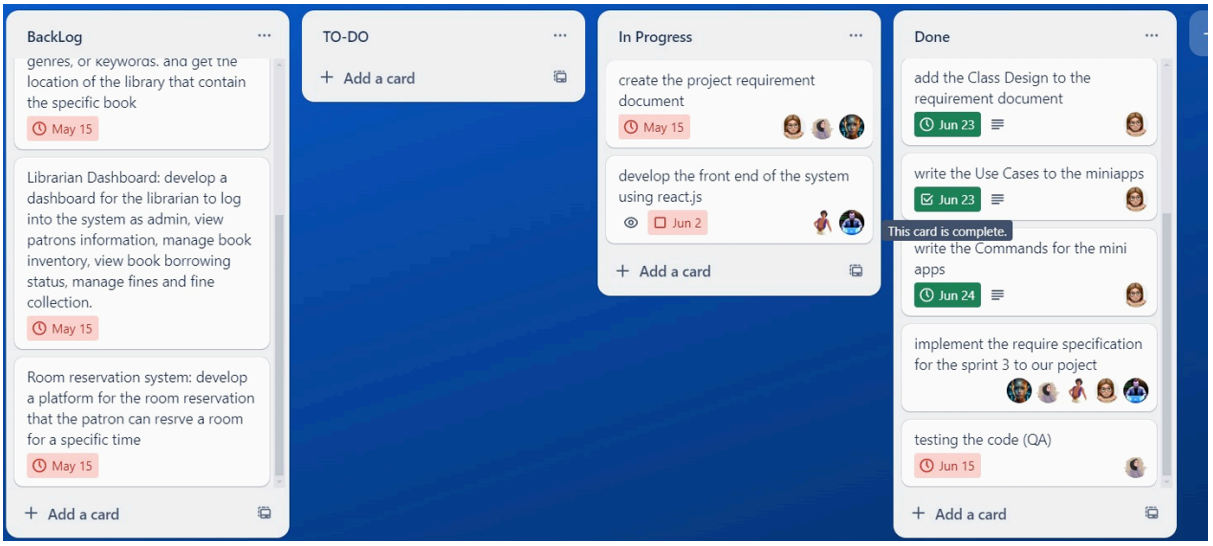
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End of the session 26.06.2024





Project General Summary

General Summary of the Project

1. **What worked well for our team throughout the semester and we would like to maintain in future industry projects**:
    - Excellent teamwork and high coordination between all team members. We all felt that we could rely on each other and that everyone's contribution was significant and valuable.
    - Effective use of project management tools like TRELLO.. This tool helped us stay organized, maintain order, and allowed us to see the project's progress at any given moment.
    - Open and transparent communication. We discussed every issue that arose, which helped us find quick solutions and prevent unnecessary conflicts.
2. **How we can improve our work in future projects**:
    - Placing greater emphasis on initial planning and setting realistic timelines. We felt that in some cases we did not accurately estimate the time required for tasks, which led to time pressure.
    - Increasing the frequency of tests to identify problems earlier.
    - Improving internal project documentation. Sometimes we found ourselves wasting time searching for information, so it's important to ensure the documentation is comprehensive and accessible to all team members.

- Similar to Sprint 2, it is recommended to break down user stories into smaller, more manageable tasks. This will ease the process and allow us to see quicker progress.
- Planning the work ahead according to the team's schedule, especially during times when a significant portion of the team is unavailable due to holidays or vacations. This will help manage unexpected situations and ensure the work progresses smoothly even with fewer available team members.

3. **What we enjoyed most about working on the project**:
    - We greatly enjoyed the process of developing advanced technological solutions. Each of us brought our knowledge and skills, and it was fascinating to see how everything came together into a functional product.
    - Working as a team was a wonderful experience. We learned a lot from each other and enjoyed the interaction and collaboration.
    - Seeing the project progress and take shape into a useful product was very satisfying. It was the moment when all our hard work paid off, giving us a sense of pride and achievement.
4. **What we would do differently if we started the project now, after the knowledge and experience gained this semester**:
    - Investing more time in initial research and architectural planning of the system. We realized that initial planning is crucial for the project's success, and it's important to dedicate the necessary time for it.
    - Dedicating time to learn new tools and technologies at the beginning of the project, rather than during it. This would have helped us work more smoothly and avoid delays.
    - Defining clearer success metrics and focusing on achieving them. This would have helped us measure our progress more accurately and ensure we were on the right track.
    - Prioritizing tasks ahead of time and planning the work so we can handle situations where we might not complete everything. It's advisable to define what can be postponed or cut if necessary.
5. **Did the team work remotely without physical meetings**:
    - Yes, part of our team work was done remotely.
        - We used tools like Microsoft Teams and Zoom and WhatsApp for daily communication. This helped us stay in touch and continue working efficiently even when we couldn't meet face-to-face.
        - We used Bitbucket for code management and collaboration. This tool was essential for version control and smooth collaboration between all team members.
6. **How did remote work affect the execution of our project**:

- Remote work made us more organized and prompted us to schedule more frequent meetings. We needed to ensure that everyone was updated and that there were no gaps in information.
- Online communication was effective, but sometimes we missed the dynamics of physical meetings. Despite this, we managed to overcome this challenge and work smoothly.
- The use of technological tools proved to be very effective and necessary for future projects. We learned to use new tools and adapt to changing situations, which strengthened our teamwork skills.

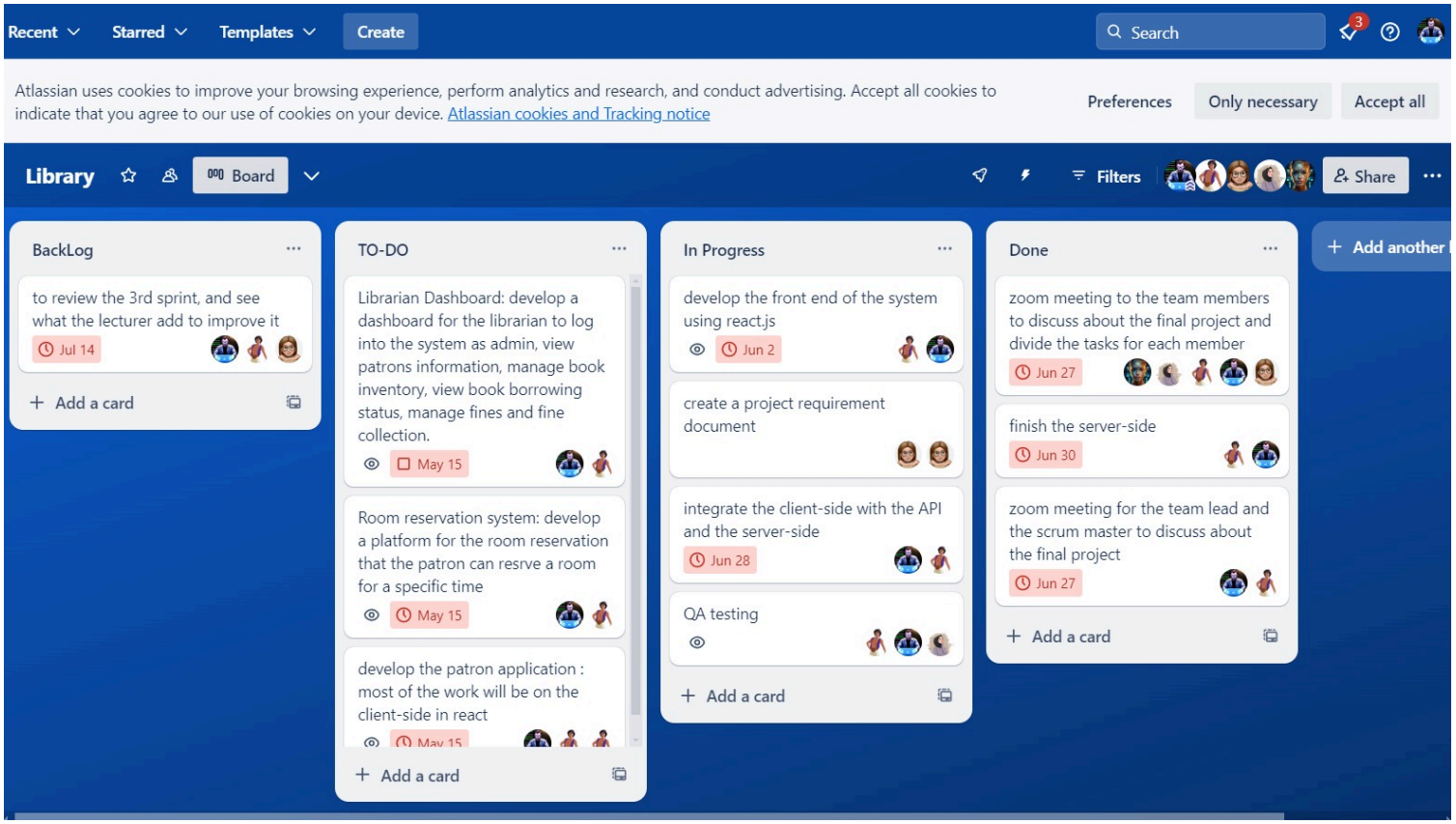
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Sprint 4
Name of project: CityLibrary

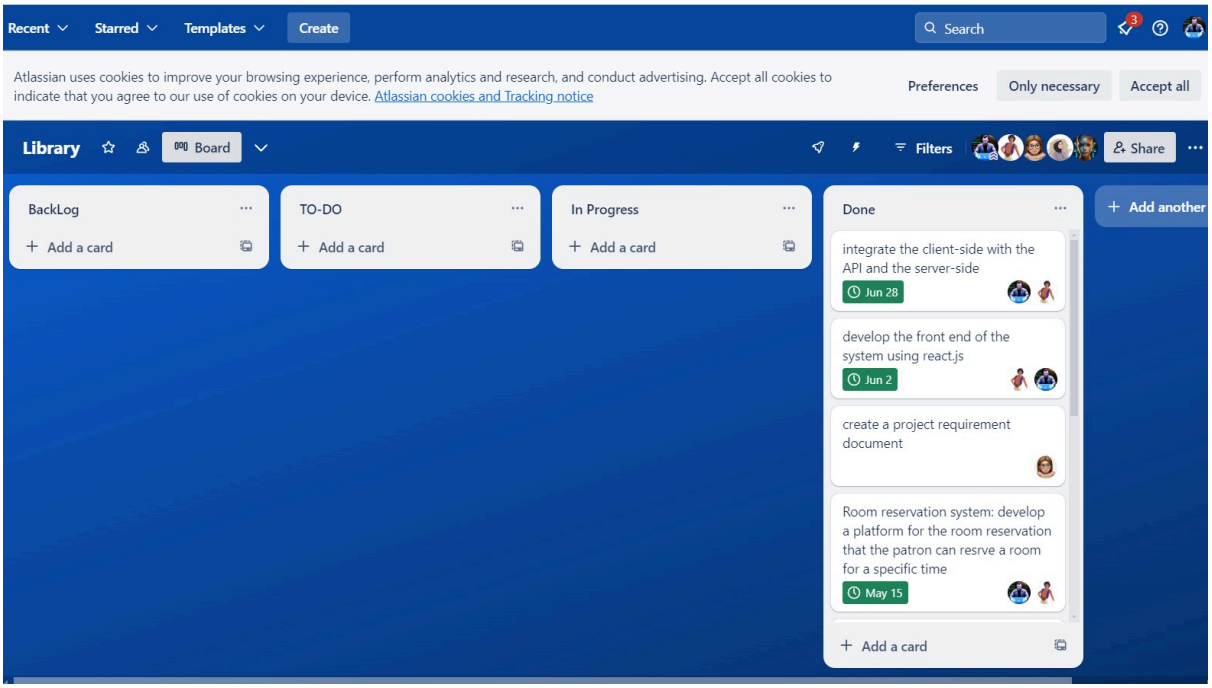
Project progress report

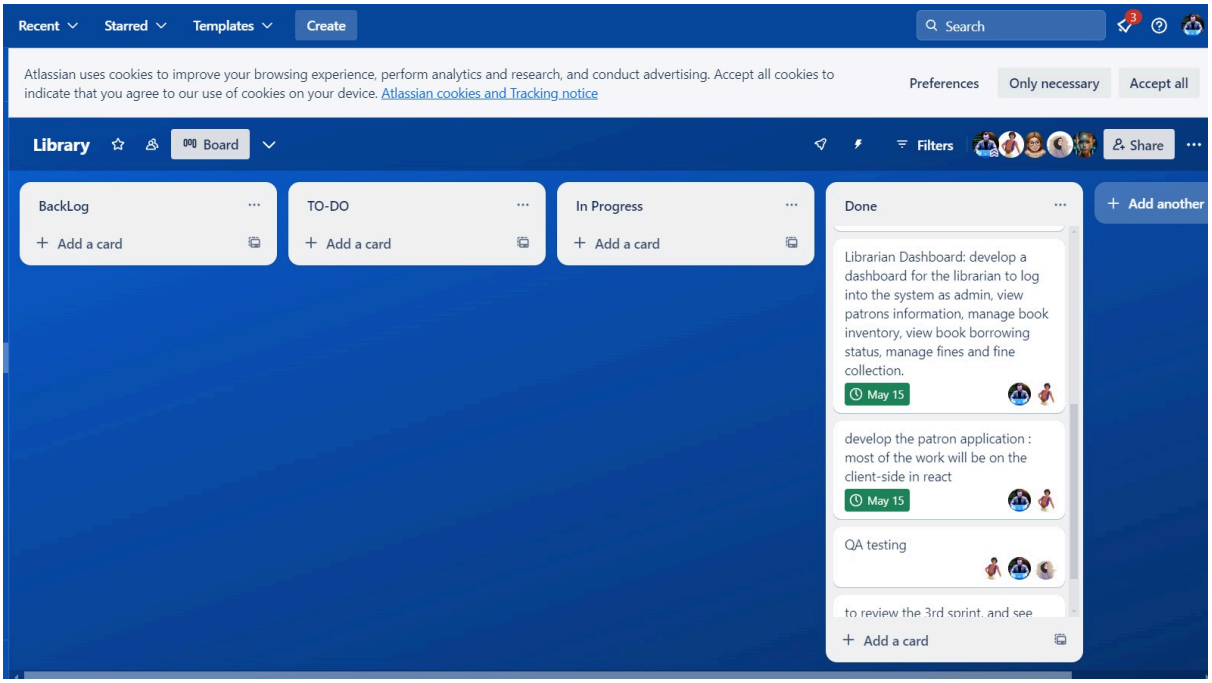
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Beginning of the session 26.6.2024



End of the session 16.7.2024





General summary of work

What went well for the team and should be continued in the next phases of work:

- Our team worked exceptionally well together. We communicated openly, shared ideas, and supported each other throughout the project. This strong teamwork should definitely be maintained in future projects.
- The use of tool like TRELLO was very effective in keeping us organized. These tools helped us visualize our progress and manage tasks efficiently.
- We were able to address and resolve issues quickly as they arose. This ability to find solutions promptly is something we should continue in future projects.
- We successfully completed the development of the server and the website, meeting our primary project goals. This demonstrates our ability to deliver results.

What should be improved in team work:

- Although we managed to complete our tasks, there were moments of time pressure due to inaccurate time estimations. Improving our planning and time estimation skills will help in future projects.
- Increasing the frequency of tests could help identify issues earlier, allowing us to address them before they become bigger problems.

- Enhancing our internal project documentation would save time and reduce confusion. Comprehensive and accessible documentation is crucial for smooth project execution and future maintenance.
- Similar to Sprint 2, it's advisable to break down user stories into smaller, more manageable tasks. This will ease the process and allow for quicker progress.
- Planning the work ahead according to the team's schedule, especially during times when a significant portion of the team is unavailable due to holidays or vacations. This will help manage unexpected situations and ensure the work progresses smoothly even with fewer available team members.

What problems did the team encounter through this phase of work:

- We faced challenges in time management, particularly in estimating how long certain tasks would take. This led to periods of intense work to meet deadlines.
- While remote work was mostly effective, there were occasional challenges with communication and coordination, which sometimes slowed down progress.
- We encountered technical difficulties during the development process, such as integration challenges and bugs. However, we were able to address these issues through collaborative problem-solving.

Why did we not complete all planned work:

- Throughout the project, we had to adjust our scope based on the time and resources available. This meant prioritizing key features and functionality while deferring some less critical tasks.
- Some unexpected challenges arose, such as technical difficulties and coordination issues, which required more time to resolve than initially anticipated.
- Limited availability of team members during certain periods, due to holidays and other commitments, impacted our ability to complete all planned work. Planning for these periods in advance could help mitigate such issues in future projects.